Elementary Mathematical Foundations of Wave Confinement Theory

Proof Compendium for the Corrected WCT Equation Registry

Richard J. Reyes

October 26, 2026

Abstract

This note collects complete proofs of standard or elementary statements associated with 25 identifiers in the corrected Wave Confinement Theory (WCT) equation registry: Sobolev embeddings and counterexamples; finite entropy and counting bounds; operator and logarithmic-flow identities; calculus of variations, dissipation, and phase-coherence estimates; and flat-torus spectral computations. Some counting and physical interpretations require independent hypotheses. Canonical statements are distinguished from auxiliary identities and the historical uncorrected formulas. The note proves neither a generic physical confinement theorem, any standard complexity-class separation, nor absence of higher-derivative ghosts.

Contents

1 Conventions **2**

1.1 Fourier transform on $\mathbb{R}^n$ . . . 2
1.2 Fourier coefficients on the flat torus . . . 2
1.3 Sobolev spaces . . . 2
1.4 The canonical corrected curvature operator . . . 2
1.5 Variation and regularity conventions . . . 3

2 Sobolev embeddings **3**

3 Spectral entropy and counting **4**

4 Operator identities **7**

5 Variational and flow lemmas **9**

6 Flat-torus eigenmodes (CLE5–CLE8) **11**

7 Registry cross-reference **13**

8 Logical scope and conclusion **14**

1 Conventions

1.1 Fourier transform on $\mathbb{R}^n$

We use the unitary Fourier transform, as in the Fourier-shell manuscript [2]:

$$\widehat{f}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-ix\cdot\xi} f(x)\, dx, \qquad f(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix\cdot\xi} \widehat{f}(\xi)\, d\xi. \tag{1}$$

Initially these hold for Schwartz functions; Fourier inversion is understood in the appropriate L^2, distributional, or absolutely convergent sense. Plancherel's theorem and the multiplier convention give

$$\|f\|_{L^2(\mathbb{R}^n)} = \left\|\widehat{f}\right\|_{L^2(\mathbb{R}^n)}, \qquad \widehat{\partial_j f} = i\xi_j \widehat{f}, \qquad \widehat{\Delta f} = -\left|\xi\right|^2 \widehat{f}. \tag{2}$$

1.2 Fourier coefficients on the flat torus

On $\mathbb{T}^n = (\mathbb{R}/2\pi\mathbb{Z})^n$ we use the *coefficient* convention of the E62 manuscript [2]:

$$\widehat{f}_k = (2\pi)^{-n} \int_{\mathbb{T}^n} f(x) e^{-ik\cdot x}\, dx, \qquad f(x) = \sum_{k\in\mathbb{Z}^n} \widehat{f}_k e^{ik\cdot x}, \tag{3}$$

so Parseval's identity is

$$\|f\|^2_{L^2(\mathbb{T}^n)} = (2\pi)^n \sum_{k\in\mathbb{Z}^n} \left|\widehat{f}_k\right|^2. \tag{4}$$

On $\prod_{j=1}^n (\mathbb{R}/L_j\mathbb{Z})$, use the phases $\exp(2\pi i \sum_j m_j x_j/L_j)$ and normalized coefficients $\left(\prod_j L_j\right)^{-1} \int f\, \overline{e_m}$.

1.3 Sobolev spaces

For $s \in \mathbb{R}$, set

$$H^s(\mathbb{R}^n) = \left\{ f \in \mathscr{S}'(\mathbb{R}^n) : \|f\|^2_{H^s} := \int_{\mathbb{R}^n} (1+|\xi|^2)^s \left|\widehat{f}(\xi)\right|^2 d\xi < \infty \right\}. \tag{5}$$

For integer $s = 2$ this agrees, up to equivalence of norms, with the weak-derivative definition involving derivatives of order at most two. C_c^∞ denotes smooth compactly supported functions.

1.4 The canonical corrected curvature operator

In accordance with M2 and E17 of the corrected registry [1], for $\varepsilon > 0$, $\alpha \in \mathbb{R}$, $z \in \mathbb{C}$ put

$$D_\varepsilon(z) = |z|^2 + \varepsilon^2 e^{-2\alpha|z|^2}, \qquad R_\varepsilon(z) = \frac{\overline{z}}{D_\varepsilon(z)}, \qquad \Theta_\varepsilon[\psi] = -(\Delta\psi) R_\varepsilon(\psi). \tag{6}$$

The denominator is strictly positive for all complex z. For fixed $z \neq 0$, $R_\varepsilon(z) \to \overline{z}/|z|^2 = 1/z$ as $\varepsilon \downarrow 0$. The earlier additive-denominator expression is a different, potentially singular object and will not be substituted for (6).

1.5 Variation and regularity conventions

All displayed integrations by parts are justified by compactly supported variations, periodic boundary conditions, or explicitly imposed vanishing boundary terms. For real-valued energies of complex fields, variations are performed in the real and imaginary parts. In a Hilbert-space gradient flow, the gradient always refers to the designated Hilbert inner product; L^2 and H^2 gradients are generally distinct.

2 Sobolev embeddings

Theorem 2.1 ([**M4/E24**] Sharp H^2 threshold)**.** *For integer $n \geq 1$,*

$$H^2(\mathbb{R}^n) \hookrightarrow L^\infty(\mathbb{R}^n) \quad \Longleftrightarrow \quad n \leq 3. \tag{7}$$

The continuous embedding means $\|f\|_\infty \leq C_n \|f\|_{H^2}$ for every $f \in H^2(\mathbb{R}^n)$.

Proof. Assume first $n < 4$. Fourier inversion and Cauchy–Schwarz give, for Schwartz f,

$$\begin{aligned} |f(x)| &\leq (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left|\widehat{f}(\xi)\right| d\xi \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} (1+|\xi|^2)^{-1}(1+|\xi|^2) \left|\widehat{f}(\xi)\right| d\xi \\ &\leq (2\pi)^{-n/2} \left(\int_{\mathbb{R}^n} (1+|\xi|^2)^{-2} \, d\xi \right)^{1/2} \|f\|_{H^2} . \end{aligned}$$

To determine the integrability of the first factor, polar coordinates yield

$$I_n = \left|\mathbb{S}^{n-1}\right| \int_0^\infty \frac{r^{n-1}}{(1+r^2)^2} \, dr.$$

Near the origin the integrand is $O(r^{n-1})$, hence integrable for $n \geq 1$. At infinity it is asymptotic to r^{n-5}, whose integral converges exactly when $n - 5 < -1$, equivalently $n < 4$. Therefore $\|f\|_\infty \leq (2\pi)^{-n/2} I_n^{1/2} \|f\|_{H^2}$. Extension by density gives the result for every $f \in H^2$, with a bounded continuous representative. The converse for $n \geq 4$ is proved by Theorem 2.3. □

Remark 2.2. The theorem concerns the *chosen* H^2 functional-analytic route. It does not show that long-lived confinement in four spatial dimensions is impossible or that every possible stability mechanism must use this embedding. On general domains Ω, extension/regularity assumptions on Ω are needed before transferring the Euclidean statement.

Theorem 2.3 ([**E67**] Explicit failure above three dimensions)**.** *For every integer $n \geq 4$ there exists $f \in H^2(\mathbb{R}^n)$ such that $f \notin L^\infty(\mathbb{R}^n)$.*

Proof. For $n = 4$, choose a radial $\chi \in C_c^\infty(\mathbb{R}^4)$ equal to one on $|x| \leq e^{-3}$ and zero on $|x| \geq e^{-2}$. For $x \neq 0$ put

$$f(x) = \chi(x) \log\log \frac{e}{|x|}.$$

Near the origin, writing $r = |x|$ and $L(r) = \log(e/r)$, we have

$$f'(r) = -\frac{1}{rL(r)}, \qquad f''(r) = \frac{L(r)-1}{r^2 L(r)^2}.$$

A radial function on $\mathbb{R}^4$ satisfies $\left|D^2 f\right|^2 = |f''|^2 + 3\,|f'/r|^2$. Consequently, for r sufficiently small,

$$\left|D^2 f(r)\right|^2 \le \frac{C}{r^4 L(r)^2}$$

and hence

$$\begin{aligned}\int_{B_{r_0}} \left|D^2 f\right|^2 dx &\le C' \int_0^{r_0} \frac{dr}{r[\log(e/r)]^2} \\ &= C' \int_{\log(e/r_0)}^{\infty} \frac{dy}{y^2} < \infty.\end{aligned}$$

The L^2 integrals of f and ∇f converge by weaker singularity estimates. The cutoff terms are smooth and supported away from the origin. Integration by parts against test functions on punctured balls, followed by removal of the inner boundary, shows that these classical derivatives agree with the distributional derivatives: the respective boundary terms vanish. Thus $f \in H^2(\mathbb{R}^4)$. But $f(r) \to +\infty$ as $r \downarrow 0$, so $f \notin L^\infty$.

For $n \ge 5$, choose

$$0 < \alpha_0 < \frac{n}{2} - 2$$

and let $f(x) = \chi(x)\,|x|^{-\alpha_0}$ with χ compactly supported and equal to one near the origin. For $j = 0, 1, 2$,

$$\left|D^j f(x)\right| \le C_j\,|x|^{-\alpha_0 - j}$$

near zero, so

$$\int_{B_{r_0}} \left|D^j f\right|^2 dx \le C \int_0^{r_0} r^{n-1-2\alpha_0-2j}\, dr < \infty.$$

The last inequality follows from $n - 2\alpha_0 - 4 > 0$. As above, the cutoff singularity has the stated weak derivatives, so $f \in H^2$. Yet $f(x) \to \infty$ at the origin. This completes both cases. □

Remark 2.4. The existence of these unbounded H^2 functions is classical and is consistent with the sharp critical Sobolev theory; see Adams–Fournier [3].

Definition 2.5 ([**E65**] Critical Sobolev power)**.** For $n > 2$ the H^1 critical Lebesgue exponent and associated semilinear power are

$$2^* = \frac{2n}{n-2}, \qquad p_c(n) := 2^* - 1 = \frac{n+2}{n-2}. \tag{8}$$

The last equality follows by subtracting one. This is a standard exponent *definition*, not a new existence or stability theorem [3, 4].

Remark 2.6. Criticality is model dependent: the displayed p_c is the usual energy-critical power for H^1-based semilinear equations, not a universally applicable H^2-critical exponent.

3 Spectral entropy and counting

Theorem 3.1 ([**E30**] Sharp entropy inequalities)**.** *If $P = (P_1, \dots, P_K)$, with $K \ge 1$, is a probability vector, $P_j \ge 0$ and $\sum_j P_j = 1$, then for $H(P) = -\sum_j P_j \ln P_j$, with $0 \ln 0 = 0$, one has*

$$0 \le H(P) \le \ln K. \tag{9}$$

The upper equality holds exactly for $P_j = 1/K$ for every j; the lower equality holds exactly for a point mass.

Proof. Since $0 \le P_j \le 1$, every term $-P_j \ln P_j$ is nonnegative. If the sum is zero, each positive P_j must equal one, and the normalization forces a point mass. For the upper bound, let $U_j = 1/K$ and use Gibbs' inequality for relative entropy:

$$0 \le D(P\|U) = \sum_{j:P_j>0} P_j \ln \frac{P_j}{U_j} = -H(P) + \ln K.$$

Thus $H(P) \le \ln K$, with equality if and only if $P = U$. Alternatively this is the strict concavity of the logarithm. □

Limitation. A finite entropy inequality does not prove monotone entropy production under an evolution. That requires a separate dynamical law.

Corollary 3.2 ([**E33**] Effective-support inequality)**.** *If $K_t \ge 1$ is the number of positive entries of a discrete probability distribution, then*

$$\boxed{e^{H(t)} \le K_t}. \tag{10}$$

Equality holds exactly for a distribution uniform on its support.

Proof. Apply Theorem 3.1 to the strictly positive entries: $H(t) \le \ln K_t$. Exponentiate both sides. □

Limitation. The inequality is a *lower* bound for actual support size; it is not an upper bound. In particular, support shrinkage is not established by the entropy inequality alone.

Theorem 3.3 ([**E29**] Cumulative exponential pruning)**.** *Suppose $M_t \ge 0$ and $\Delta_t \ge 0$ obey $M_{t+1} \le e^{-\Delta_t} M_t$. Then for every integer $T \ge 1$,*

$$\boxed{M_T \le M_0 \exp\left(-\sum_{t=0}^{T-1} \Delta_t\right)}. \tag{11}$$

Proof. At $T = 1$, (11) is exactly the given recurrence. Assume it holds at T. Then

$$\begin{aligned} M_{T+1} &\le e^{-\Delta_T} M_T \\ &\le e^{-\Delta_T} M_0 \exp\left(-\sum_{t=0}^{T-1} \Delta_t\right) = M_0 \exp\left(-\sum_{t=0}^{T} \Delta_t\right). \end{aligned}$$

The result follows by induction. □

Corollary 3.4 (Fixed fraction)**.** *If $0 \le \delta \le 1$ and $M_{t+1} \le (1-\delta) M_t$, then*

$$M_T \le (1-\delta)^T M_0.$$

Proof. The claim follows by the same induction. For $\delta < 1$ it is Theorem 3.3 with $\Delta_t = -\ln(1-\delta)$. At $\delta = 1$, $M_t = 0$ for $t \ge 1$. □

Limitation. The *canonical* E29 hypothesis is the exponential factor $e^{-\Delta_t}$, not an assumed constant retention fraction. Entropy reduction does not imply the recurrence without a model-specific bridge.

Theorem 3.5 ([**E28**] Corrected alpha-drop identity)**.** *Let $n \in \mathbb{N}$, $m(n) \in \mathbb{N} \cup \{0\}$, $\rho_t(n) \in (0,1]$ and $\beta(n) \in \mathbb{R}$, and define*

$$\alpha(n) := 1 + \frac{1}{n} \sum_{t=1}^{m(n)} \log_2 \rho_t(n) + \beta(n). \tag{12}$$

Then

$$\alpha(n) < 1 \quad \Longleftrightarrow \quad \beta(n) < -\frac{1}{n} \sum_{t=1}^{m(n)} \log_2 \rho_t(n). \tag{13}$$

Proof. Subtract one from both sides of $\alpha(n) < 1$, substitute (12), and rearrange. Each step is reversible. □

Remark 3.6. The retained-fraction factors $\rho_t \in (0,1]$ replace the historical ratio $(M_t + 1)/M_t > 1$, which cannot by itself induce an exponent drop. The formula does not show that a proposed physical or algorithmic filter actually attains those fractions.

Theorem 3.7 ([**E32**] Uniform exponent margin)**.** *Suppose there exist $n_0 \geq 1$ and $\eta > 0$ such that for every $n \geq n_0$,*

$$\beta(n) + \frac{1}{n} \sum_{t=1}^{m(n)} \log_2 \rho_t(n) \leq -\eta. \tag{14}$$

Then

$$\boxed{\limsup_{n \to \infty} \alpha(n) \leq 1 - \eta < 1}.$$

Proof. By (12) and (14), $\alpha(n) \leq 1 - \eta$ for all $n \geq n_0$. Taking the limit superior establishes the bound. □

Remark 3.8 (Terminology). A bound of order $2^{(1-\eta)n}$ with $0 < \eta < 1$ is still *exponential* in n, though its exponent improves on 2^n. The usual asymptotic meaning of *subexponential* is $2^{o(n)}$. The canonical registry title "Subexponential exploration condition" should not be interpreted as proving $2^{o(n)}$ from $\limsup \alpha < 1$.

Theorem 3.9 ([**E41**] Conditional configuration count)**.** *Let an initial finite binary code set $B_0(n)$ satisfy $|B_0(n)| \leq 2^n$. Assume a sequence of finite retained sets $B_t(n)$ satisfies*

$$|B_t(n)| \leq \rho_t(n)\, |B_{t-1}(n)|\,, \qquad \rho_t(n) \in (0,1]. \tag{15}$$

Suppose a finite code set A_n has cardinality

$$|A_n| \leq 2^{n\beta(n)} \left| B_{m(n)}(n) \right| \tag{16}$$

and the physically defined set $C_{\mathrm{curv}}(n)$ admits an injection into A_n. Then

$$\boxed{|C_{\mathrm{curv}}(n)| \leq 2^{n\alpha(n)}}. \tag{17}$$

Proof. Repeated application of (15) gives $|B_m| \leq |B_0| \prod_{t=1}^{m} \rho_t(n)$. An injection cannot increase cardinality, so

$$\begin{aligned} |C_{\mathrm{curv}}(n)| \leq |A_n| &\leq 2^{n\beta(n)}\, |B_0(n)| \prod_{t=1}^{m(n)} \rho_t(n) \\ &\leq 2^{n + n\beta(n) + \sum_{t=1}^{m(n)} \log_2 \rho_t(n)} = 2^{n\alpha(n)}. \end{aligned}$$

□

Limitation. The injections and cardinality retention inequalities are *hypotheses*, not consequences of E28 alone. If the right side is less than one, the hypotheses force the discrete retained set to be empty. No polynomial-time enumeration follows from (17).

Corollary 3.10 ([**E72**] Effective search-space count)**.** *If the effective search space $S_{\text{eff}}(n)$ admits an injection into a finite code set A_n satisfying the cardinality hypotheses of Theorem 3.9, then*

$$\boxed{|S_{\text{eff}}(n)| \le 2^{n\alpha(n)}}.$$

Proof. Injectivity yields $|S_{\text{eff}}(n)| \le |A_n|$; apply the bound on A_n in Theorem 3.9. □

Limitation. Neither E41 nor E72 proves any separation between standard complexity classes. In particular, a configuration count alone does not prove a computationally efficient encoding or pruning algorithm.

4 Operator identities

Lemma 4.1 (Auxiliary Laplacian-multiplication commutator)**.** *Let $f, v \in C^2(\Omega)$ be scalar-valued. With $[\Delta, f]v := \Delta(fv) - f\Delta v$,*

$$\boxed{[\Delta, f]v = (\Delta f)v + 2\nabla f \cdot \nabla v}. \tag{18}$$

Proof. Twice applying the product rule yields

$$\partial_j^2(fv) = (\partial_j^2 f)v + 2(\partial_j f)(\partial_j v) + f\partial_j^2 v.$$

Sum over j and subtract $f\Delta v$. □

Remark 4.2. This is a useful auxiliary identity, but it is *not* the canonical E51 operator commutator, which is proved next.

Theorem 4.3 ([**E51**] Frozen-denominator curvature commutator)**.** *Let $D \in C^1(\Omega; \mathbb{C})$ have no zero, let $\psi \in C^3(\Omega; \mathbb{C})$, and define a* linear frozen-coefficient *operator $\Theta_D[v] := -(\Delta v)/D$. Applying it componentwise to a vector field,*

$$\boxed{\nabla(\Theta_D[\psi]) - \Theta_D[\nabla\psi] = \frac{\Delta\psi}{D^2}\nabla D}. \tag{19}$$

Proof. By the quotient rule,

$$\nabla\left(-\frac{\Delta\psi}{D}\right) = -\frac{\nabla\Delta\psi}{D} + \frac{\Delta\psi}{D^2}\nabla D.$$

Constant-coefficient partial derivatives commute on $\mathbb{R}^n$, so $\nabla\Delta\psi = \Delta\nabla\psi$. The first term is $\Theta_D[\nabla\psi]$; subtract it to obtain (19). □

Remark 4.4 (Relation to the corrected curvature operator). The true corrected map $\Theta_\varepsilon[\psi] = -(\Delta\psi)R_\varepsilon(\psi)$ is nonlinear because its multiplier depends on ψ and $\overline{\psi}$. If one freezes the coefficient at a chosen field and defines $T_\psi[v] = -R_\varepsilon(\psi)\Delta v$, then the exact related identity is

$$\nabla(T_\psi\psi) - T_\psi(\nabla\psi) = -(\Delta\psi)\nabla(R_\varepsilon(\psi)).$$

Neither formula permits replacing the frozen coefficient by one evaluated afresh on the vector field $\nabla\psi$.

Theorem 4.5 ([**EX**] Logarithmic Laplacian identity)**.** *Let $u \in C^2(\Omega;\mathbb{R})$ and $\psi = e^u > 0$. Then*

$$\boxed{\frac{\Delta\psi}{\psi} = \Delta u + |\nabla u|^2}. \tag{20}$$

Proof. The chain rule gives $\nabla\psi = e^u \nabla u$. Therefore

$$\Delta\psi = \nabla \cdot (e^u \nabla u) = e^u \, |\nabla u|^2 + e^u \Delta u.$$

Divide by $e^u > 0$. □

Theorem 4.6 ([**EY/EZ**] Cole–Hopf/logarithmic reduction)**.** *Let $u \in C^{2,1}(\Omega \times (0,T);\mathbb{R})$ satisfy*

$$\partial_t u = \Delta u + |\nabla u|^2 . \tag{21}$$

Then $\psi = e^u$ satisfies the heat equation

$$\boxed{\partial_t \psi = \Delta \psi}. \tag{22}$$

Conversely, every strictly positive $C^{2,1}$ solution of (22) gives a solution of (21) by $u = \ln \psi$.

Proof. Differentiation and Theorem 4.5 show

$$\psi_t = e^u u_t = e^u(\Delta u + |\nabla u|^2) = \Delta(e^u).$$

Conversely, positivity permits $u = \ln \psi$ and hence

$$u_t = \frac{\psi_t}{\psi} = \frac{\Delta\psi}{\psi} = \Delta u + |\nabla u|^2 .$$

□

Remark 4.7. This is the classical logarithmic version of the Cole–Hopf transformation; see Evans [4, §4.4]. It does not globally linearize an arbitrary WCT evolution; the real logarithm breaks down at zeros or sign changes of the field.

Theorem 4.8 ([**E9**] Polar phase-flux identity)**.** *Suppose ψ is a differentiable complex field and locally, on $\{\psi \neq 0\}$, write $\psi = \sqrt{\rho}\, e^{i\vartheta}$ with $\rho = |\psi|^2$. Then*

$$\boxed{\mathrm{Im}(\overline{\psi}\nabla\psi) = \rho\nabla\vartheta}. \tag{23}$$

Proof. Differentiating the polar representation yields

$$\nabla\psi = e^{i\vartheta}\nabla\sqrt{\rho} + i\sqrt{\rho}\, e^{i\vartheta}\nabla\vartheta.$$

Multiplying by $\overline{\psi} = \sqrt{\rho}\, e^{-i\vartheta}$ gives

$$\overline{\psi}\nabla\psi = \sqrt{\rho}\,\nabla\sqrt{\rho} + i\rho\nabla\vartheta.$$

The first term is real, so taking the imaginary part proves (23). □

Limitation. This is a kinematic identity. A continuity law $\rho_t + \nabla \cdot \mathbf{S} = 0$ needs an additional field equation or conservation principle; it does not follow from (23).

5 Variational and flow lemmas

Theorem 5.1 ([**E20**] Quadratic-sector positivity)**.** *For $\kappa, \theta, \gamma \in \mathbb{R}$ set $Q(a,b) = \kappa a^2 + \gamma ab + \theta b^2$. Then*

$$\boxed{Q(a,b) \geq 0 \quad \textit{for all } a,b \in \mathbb{R} \iff \kappa \geq 0,\ \theta \geq 0,\ \gamma^2 \leq 4\kappa\theta}. \tag{24}$$

Proof. Taking $(a,b) = (1,0)$ and $(0,1)$ gives $\kappa, \theta \geq 0$. If $\kappa > 0$, complete the square:

$$Q(a,b) = \kappa \left(a + \frac{\gamma}{2\kappa} b\right)^2 + \left(\theta - \frac{\gamma^2}{4\kappa}\right) b^2.$$

Nonnegativity for every (a,b) requires $\gamma^2 \leq 4\kappa\theta$. If $\kappa = 0$, $Q(a,b) = \gamma ab + \theta b^2$ can be nonnegative for all (a,b) only if $\gamma = 0$, which is the same condition. Conversely, the completed-square expression is nonnegative whenever $\kappa > 0$ and the three inequalities hold. If $\kappa = 0$, those inequalities imply $\gamma = 0$, so $Q = \theta b^2 \geq 0$. □

Remark 5.2 (Sign convention and ghost obstruction). The canonical E20 sector is $Q(S,P) = \kappa S^2 + \theta P^2 - \gamma SP$. The criterion is unchanged on sending $\gamma \mapsto -\gamma$. Positivity of this algebraic quadratic sector does *not* establish boundedness of the canonical Hamiltonian, reduction of higher-time configuration degrees of freedom, or absence of an Ostrogradsky ghost. The nondegenerate fixed-background S004 analysis is a separate canonical-constraint result [11].

Theorem 5.3 ([**E21**] Second-derivative Euler–Lagrange equation)**.** *Let*

$$S[u] = \int_\Omega L(u, \partial_\mu u, \partial_\mu \partial_\nu u)\, dx$$

with enough differentiability to justify first variation and integration by parts twice. If u is stationary for all compactly supported smooth variations, then

$$\boxed{\frac{\partial L}{\partial u} - \partial_\mu \left(\frac{\partial L}{\partial(\partial_\mu u)}\right) + \partial_\mu \partial_\nu \left(\frac{\partial L}{\partial(\partial_\mu \partial_\nu u)}\right) = 0}. \tag{25}$$

Proof. For $v \in C_c^\infty(\Omega)$ consider $u_\varepsilon = u + \varepsilon v$. The first variation is

$$\frac{d}{d\varepsilon} S[u_\varepsilon]\Big|_{\varepsilon=0} = \int_\Omega \left[\frac{\partial L}{\partial u} v + \frac{\partial L}{\partial(\partial_\mu u)} \partial_\mu v + \frac{\partial L}{\partial(\partial_\mu \partial_\nu u)} \partial_\mu \partial_\nu v\right] dx.$$

Integrating the first derivative term by parts once and the second-derivative term by parts twice, with vanishing boundary terms, produces

$$\delta S[u](v) = \int_\Omega \left[\frac{\partial L}{\partial u} - \partial_\mu \left(\frac{\partial L}{\partial(\partial_\mu u)}\right) + \partial_\mu \partial_\nu \left(\frac{\partial L}{\partial(\partial_\mu \partial_\nu u)}\right)\right] v\, dx.$$

This vanishes for arbitrary v. The fundamental lemma of the calculus of variations gives (25) (distributionally, and classically when the indicated derivatives exist). □

Limitation. An Euler–Lagrange equation does not by itself establish that its Cauchy problem is well posed or ghost free.

Theorem 5.4 ([**TOP3**] Hilbert-space gradient dissipation)**.** *Let $\mathcal{H}$ be a real Hilbert space, $E : \mathcal{H} \to \mathbb{R}$ continuously Fréchet differentiable, and $u(t)$ a sufficiently regular solution of*

$$u_t = -\nabla_{\mathcal{H}} E(u).$$

Then

$$\boxed{\frac{d}{dt} E(u(t)) = -\left\|\nabla_{\mathcal{H}} E(u(t))\right\|_{\mathcal{H}}^2 \leq 0}. \tag{26}$$

For all times T on which the solution and chain rule exist,

$$E(u(T)) + \int_0^T \left\|\nabla_{\mathcal{H}} E(u(t))\right\|_{\mathcal{H}}^2 \, dt = E(u(0)). \tag{27}$$

Proof. The chain rule and Riesz representation yield

$$\begin{aligned} \frac{dE}{dt} &= DE(u)[u_t] = \langle \nabla_{\mathcal{H}} E(u), u_t \rangle_{\mathcal{H}} \\ &= -\langle \nabla_{\mathcal{H}} E(u), \nabla_{\mathcal{H}} E(u) \rangle_{\mathcal{H}} = -\left\|\nabla_{\mathcal{H}} E(u)\right\|_{\mathcal{H}}^2 . \end{aligned}$$

Integrating this identity over $[0, T]$ proves (27). □

Remark 5.5. Canonical TOP3 applies this principle to an energy of curves γ. To do that rigorously, one must specify the curve-space metric, parametrization, regularity, and treatment of self-avoidance. Energy descent alone does not prove global existence, unique equilibrium, or convergence to a desired geometry.

Theorem 5.6 ([**E50**] Phase-coherence integrability)**.** *Let $\psi \in L^2(\Omega; \mathbb{C})$ and suppose a measurable weak phase gradient $\nabla\vartheta$ is defined where $\psi \neq 0$ and satisfies $|\nabla\vartheta| \geq \delta > 0$ almost everywhere on that region. Define the integrand to be zero on $\{\psi = 0\}$. Then*

$$\mathcal{C}[\psi] = \int_\Omega \frac{|\psi|^2}{|\nabla\vartheta|} \, dx, \qquad \boxed{0 \leq \mathcal{C}[\psi] \leq \delta^{-1} \left\|\psi\right\|_{L^2}^2 < \infty}. \tag{28}$$

Proof. The gradient-floor assumption gives almost everywhere

$$0 \leq \frac{|\psi|^2}{|\nabla\vartheta|} \leq \frac{|\psi|^2}{\delta}.$$

Integrating and using $\psi \in L^2$ proves the bound. □

Remark 5.7. Alternatively the globally regularized denominator $\sqrt{|\nabla\vartheta|^2 + \delta^2}$ yields the same estimate where its gradient is defined. Neither formulation proves that a WCT flow generates or preserves the required phase-gradient lower bound.

Theorem 5.8 ([**CM9**] First-order mode-system equivalence)**.** *Let $\delta_\gamma, \delta_b \in C^2(I)$ and suppose the prescribed second-order system is*

$$\ddot{\delta}_\gamma = -c_s^2 k^2 \delta_\gamma - k^2 \Phi, \qquad \ddot{\delta}_b = -\mathcal{R} c_s^2 k^2 \delta_\gamma - k^2 \Phi.$$

Set $v_\gamma = \dot{\delta}_\gamma$ and $v_b = \dot{\delta}_b$. Then the system is equivalent, with matching initial positions and velocities, to

$$\boxed{\begin{aligned} \dot{\delta}_\gamma &= v_\gamma, \quad \dot{v}_\gamma = -c_s^2 k^2 \delta_\gamma - k^2 \Phi, \\ \dot{\delta}_b &= v_b, \quad \dot{v}_b = -\mathcal{R} c_s^2 k^2 \delta_\gamma - k^2 \Phi. \end{aligned}} \tag{29}$$

Proof. Differentiating the definitions of v_γ, v_b and using the second-order equations yields (29). Conversely, differentiating the first and third equations in (29) and substituting the second and fourth recovers the second-order system. Matching initial data makes these maps inverses. □

Limitation. This is an ODE equivalence, not a derivation of cosmological coefficients, gravitational forcing, or predictive physical dynamics.

6 Flat-torus eigenmodes (CLE5–CLE8)

Theorem 6.1 ([**CLE5**] Flat product-torus eigenfunctions)**.** *Let*

$$\mathbb{T}^2_{L_1,L_2} = (\mathbb{R}/L_1\mathbb{Z}) \times (\mathbb{R}/L_2\mathbb{Z}), \qquad L_1, L_2 > 0,$$

with its flat product metric. For $m, n \in \mathbb{Z}$,

$$\varphi_{m,n}(s,t) = \exp\left(2\pi i \left(\frac{ms}{L_1} + \frac{nt}{L_2}\right)\right) \tag{30}$$

is an eigenfunction of $-\Delta$, with

$$\boxed{\lambda_{m,n} = \left(\frac{2\pi m}{L_1}\right)^2 + \left(\frac{2\pi n}{L_2}\right)^2}. \tag{31}$$

The modes constitute a complete orthogonal Fourier basis of L^2.

Proof. Direct differentiation gives

$$\partial_s^2 \varphi_{m,n} = -\left(\frac{2\pi m}{L_1}\right)^2 \varphi_{m,n}, \quad \partial_t^2 \varphi_{m,n} = -\left(\frac{2\pi n}{L_2}\right)^2 \varphi_{m,n}.$$

Since $\Delta = \partial_s^2 + \partial_t^2$ for the flat metric, adding proves (31). Integer m, n give periodicity, and completeness is the product Fourier series theorem. □

Lemma 6.2 ([**CLE6**] Separation in angular coordinates)**.** *On the flat angular torus with metric $R^2 d\theta^2 + r^2 d\phi^2$, let $-\Delta\psi = \sigma_\star^2 \psi$ and suppose $\psi(\theta,\phi) = f(\theta)g(\phi)$. Then, wherever $fg \neq 0$,*

$$\boxed{\frac{f''}{f} + \frac{R^2}{r^2}\frac{g''}{g} = -\sigma_\star^2 R^2}. \tag{32}$$

Proof. The Laplacian is $R^{-2}\partial_\theta^2 + r^{-2}\partial_\phi^2$. Insert $\psi = fg$ into the eigenvalue equation to obtain $-R^{-2}f''g - r^{-2}fg'' = \sigma_\star^2 fg$. Dividing by fg and multiplying by $-R^2$ yields (32). The undivided equation remains valid at zeros. □

Theorem 6.3 ([**CLE7**] Periodic angular ODE)**.** *For integer $m \geq 0$, every 2π-periodic classical solution of $f'' + m^2 f = 0$ is*

$$\boxed{f(\theta) = A\cos(m\theta) + B\sin(m\theta)}. \tag{33}$$

If $m = 0$, periodicity forces the solution to be constant.

Proof. For $m > 0$, the characteristic roots are $\pm im$, yielding exactly the two-dimensional solution space (33); integer m ensures periodicity. For $m = 0$, the general solution is $A + B\theta$, and periodicity requires $B = 0$. □

Theorem 6.4 (Lowest nonzero flat-torus eigenvalue)**.** *If $L_1 > L_2 > 0$, the least positive eigenvalue of $-\Delta$ on $\mathbb{T}^2_{L_1,L_2}$ is*

$$\boxed{\lambda_1 = \left(\frac{2\pi}{L_1}\right)^2},$$

attained precisely for $(m, n) = (\pm 1, 0)$.

Proof. For every nonzero integer pair (m, n), either $m \neq 0$ or $n \neq 0$. In the first case, $\lambda_{m,n} \geq (2\pi/L_1)^2$, with equality only for $|m| = 1$ and $n = 0$. In the second case with $m = 0$, $\lambda_{0,n} \geq (2\pi/L_2)^2 > (2\pi/L_1)^2$. Thus precisely the indicated two modes realize the least positive eigenvalue. □

Theorem 6.5 (**[CLE8]** Fixed-winding spectral selection)**.** *On the flat angular torus with $R, r > 0$, $\psi(\theta, \phi) = Ae^{i\phi}$ is an eigenfunction with eigenvalue r^{-2}. In the sector of separated Fourier modes $e^{i(m\theta+\phi)}$ with fixed ϕ-winding $n = 1$, this is the minimum eigenvalue, achieved exactly when $m = 0$.*

Proof. The flat product spectrum in angular coordinates is

$$\lambda_{m,n} = \frac{m^2}{R^2} + \frac{n^2}{r^2}.$$

At fixed $n = 1$,

$$\lambda_{m,1} = \frac{m^2}{R^2} + \frac{1}{r^2} \geq \frac{1}{r^2},$$

with equality only for $m = 0$. The mode $Ae^{i\phi}$ is exactly $(m, n) = (0, 1)$. □

Remark 6.6 (Global versus fixed-winding selection). If $R > r$, then the global lowest nonzero angular mode is $e^{\pm i\theta}$, with eigenvalue $R^{-2} < r^{-2}$. Thus canonical CLE8 does not claim the global lowest nonzero product-torus eigenvalue: its minimization is within a fixed winding sector. Normalization, chirality, and physical electron identification require additional assumptions.

Theorem 6.7 (Quantified embedded-torus operator error)**.** *Let $R > r > 0$, and consider the standard torus embedded in $\mathbb{R}^3$:*

$$X(\theta, \phi) = \big((R + r\cos\phi)\cos\theta, (R + r\cos\phi)\sin\theta, r\sin\phi\big).$$

Its Laplace–Beltrami operator and the flat product comparison are

$$\Delta_g v = \frac{\partial_\theta^2 v}{(R + r\cos\phi)^2} + \frac{\partial_\phi^2 v}{r^2} - \frac{\sin\phi}{r(R + r\cos\phi)}\partial_\phi v, \quad \Delta_{\text{flat}} v = \frac{1}{R^2}\partial_\theta^2 v + \frac{1}{r^2}\partial_\phi^2 v. \tag{34}$$

For $v \in C^2(\mathbb{T}^2)$ the absolute operator error satisfies

$$\boxed{\begin{aligned}\|(\Delta_g - \Delta_{\text{flat}})v\|_\infty \leq \frac{2Rr + r^2}{R^2(R - r)^2}\left\|\partial_\theta^2 v\right\|_\infty \\ + \frac{1}{r(R - r)}\left\|\partial_\phi v\right\|_\infty.\end{aligned}} \tag{35}$$

Proof. The induced metric is diagonal: $g_{\theta\theta} = (R + r\cos\phi)^2$, $g_{\phi\phi} = r^2$, and $\sqrt{\det g} = r(R + r\cos\phi)$. Substitute into

$$\Delta_g v = \frac{1}{\sqrt{\det g}} \partial_i \big(\sqrt{\det g}\, g^{ij} \partial_j v \big)$$

to obtain (34). Subtracting Δ_{flat} gives

$$\begin{aligned}(\Delta_g - \Delta_{\text{flat}})v = &\left[\frac{1}{(R + r\cos\phi)^2} - \frac{1}{R^2} \right] \partial_\theta^2 v \\ &- \frac{\sin\phi}{r(R + r\cos\phi)} \partial_\phi v.\end{aligned}$$

Since $R + r\cos\phi \geq R - r > 0$,

$$\left| \frac{1}{(R + r\cos\phi)^2} - \frac{1}{R^2} \right| \leq \frac{2Rr + r^2}{R^2(R - r)^2}, \quad \left| \frac{\sin\phi}{r(R + r\cos\phi)} \right| \leq \frac{1}{r(R - r)}.$$

Multiply these bounds by the indicated derivative norms and take the supremum to obtain (35). □

Remark 6.8. The flat product operator is exact only for the *flat product* metric. On an embedded torus the metric-dependent first derivative and coefficients must be retained or controlled. The displayed error is a function-dependent absolute estimate, not a uniform relative eigenvalue-error theorem, and "thin torus" alone should not be treated as zero error for all families of high-frequency modes.

7 Registry cross-reference

The table distinguishes theorem type and specific WCT source family. A manuscript reference identifies relevant context, not a claim that its full physical theory is proved by the elementary lemma.

ID	Result	Status	Supporting full paper or source
M4	Thm. 2.1	Standard	*Hard Upper Bound on Spatial Dimensionality in WCT* [5]
E24	Thm. 2.1	Standard	*Hard Upper Bound on Spatial Dimensionality in WCT* [5]
E67	Thm. 2.3	Standard counterexample	Dimensionality manuscript; *Sobolev Spaces* [5, 3]
E65	Def. 2.5	Standard definition	Dimensionality manuscript and standard Sobolev theory [5, 3]
E30	Thm. 3.1	Standard	*Self-Emergent Fourier Cymatics* [6]
E33	Cor. 3.2	Elementary	*Self-Emergent Fourier Cymatics* [6]
E29	Thm. 3.3	Elementary	Corrected WCT equation registry, entropy/counting section [1]
E28	Thm. 3.5	Elementary	*P vs NP in Curvature-Bounded Wave Computation* [7]
E32	Thm. 3.7	Conditional limit	*P vs NP in Curvature-Bounded Wave Computation* [7]
E41	Thm. 3.9	Conditional counting	*P vs NP in Curvature-Bounded Wave Computation* [7]
E72	Cor. 3.10	Conditional counting	*P vs NP in Curvature-Bounded Wave Computation* [7]
E51	Thm. 4.3	Elementary identity	*The Geometry of Resonance*; corrected registry [10, 1]
EX	Thm. 4.5	Standard	*Logarithmic Curvature Flow* [9]
EY	Thm. 4.6	Standard	*Logarithmic Curvature Flow* [9]

ID	Result	Status	Supporting full paper or source
EZ	Thm. 4.6	Standard	*Logarithmic Curvature Flow* [9]
E9	Thm. 4.8	Elementary identity	*Phase-Flux Field* [8]
E20	Thm. 5.1	Elementary algebra	*The Geometry of Resonance*; S004 analysis [10, 11]
E21	Thm. 5.3	Standard variation	*The Geometry of Resonance* [10]
TOP3	Thm. 5.4	Standard, conditional flow	Corrected WCT registry, topology module [1]
E50	Thm. 5.6	Conditional integral	Corrected WCT registry, coherence module [1]
CM9	Thm. 5.8	Elementary ODE	Corrected WCT registry, cosmology module [1]
CLE5	Thms. 6.1, 6.7	Exact/approximate	Corrected WCT registry, curvature-locked electron module [1]
CLE6	Lem. 6.2	Standard	Corrected WCT registry, curvature-locked electron module [1]
CLE7	Thm. 6.3	Standard	Corrected WCT registry, curvature-locked electron module [1]
CLE8	Thms. 6.4, 6.5	Conditional selection	Corrected WCT registry, curvature-locked electron module [1]

8 Logical scope and conclusion

The fully analytic/elementary implications in this compendium are

(i) the sharp $H^2 \to L^\infty$ dimension threshold and explicit counterexamples above three dimensions;

(ii) entropy and exponential-iteration bounds, with precise equality conditions or recurrences;

(iii) standard differential, logarithmic, phase and commutator identities;

(iv) quadratic-form positivity, Euler–Lagrange and exact negative-gradient-flow dissipation formulas; and

(v) the exact flat product-torus spectrum, plus an explicit embedded-torus operator-error estimate.

The exponent-margin conclusion E32 and the discrete bounds E41/E72 require independent uniform retention/counting hypotheses. The phase coherence estimate E50 requires a phase-gradient floor. None of those hypotheses is obtained merely by assigning PASS status to a symbolic registry check.

Crucially, these proofs do *not* show that $n = 4$ confinement is physically impossible; that a positive algebraic quadratic sector avoids Ostrogradsky ghosts; that entropy control implies actual dynamical spectral concentration; that a bounded code count implies a complexity separation; or that an admissible winding-one toroidal mode is a unique physical particle. Such statements require separate theorems and data.

References

[1] R. J. Reyes, *Wave Confinement Theory—Corrected Full Equation Registry*, revision 2.0, June 27, 2026 (with later scope notices). https://github.com/rickyjreyes/geometry_of_resonance/blob/main/WCT_FULL_EQUATION_LIST_CORRECTED.md.

[2] R. J. Reyes, *Mathematical Manuscripts and Formal Proof Artifacts*, versioned research repository, including E62 and Fourier-shell manuscripts. https://github.com/rickyjreyes/math.

[3] R. A. Adams and J. J. F. Fournier, *Sobolev Spaces*, 2nd ed., Pure and Applied Mathematics 140, Academic Press, 2003.

[4] L. C. Evans, *Partial Differential Equations*, 2nd ed., Graduate Studies in Mathematics 19, American Mathematical Society, 2010.

[5] R. J. Reyes, *Hard Upper Bound on Spatial Dimensionality in Wave Confinement Theory*, Zenodo, 2025. https://doi.org/10.5281/zenodo.17081283.

[6] R. J. Reyes, *Self-Emergent Fourier Cymatics: Entropic Eigenmodes out of Chaos*, Zenodo, 2025. https://doi.org/10.5281/zenodo.17732648.

[7] R. J. Reyes, *P vs NP in Curvature-Bounded Wave Computation: A Model-Relative Separation*, Zenodo, 2025. https://doi.org/10.5281/zenodo.17743607.

[8] R. J. Reyes, *Phase-Flux Field (PFF): Axiomatic Substrate for Wave Confinement Theory*, Zenodo, 2025. https://doi.org/10.5281/zenodo.17578766.

[9] R. J. Reyes, *Logarithmic Curvature Flow, Filament Localization, and the Geometric Origin of the Lepton Mass Spectrum*, Zenodo, 2026. https://doi.org/10.5281/zenodo.18936949.

[10] R. J. Reyes, *The Geometry of Resonance: Wave Confinement Theory and the Emergence of Mass, Force, and Spacetime*, Zenodo, 2025. https://doi.org/10.5281/zenodo.15644222.

[11] R. J. Reyes, *Ghost and Constraint Analysis: Corrected S004 Sector*, foundational technical note, 2026. https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/foundational/GHOST_AND_CONSTRAINT_ANALYSIS.md.